Aims and objectives:

60 to 90 second 2D animations (1920x1080, 16:9 HD ratio) for the following topics:

1. What are stem cells specific focus on dental pulp stem cells.
2. Why Baby Teeth are valuable for stem cell therapy compared to adult teeth.
3. How the collection process works for tooth 4 life?

Content for Video 1

Introductory phrase: "What are stem cells"

Animation A human body slowly rotates in the centre of the screen. As it turns, the animation highlights different body parts one by one, soft glowing circles appear over the head, heart, bones, muscles, and skin. The camera then zooms in toward the mouth, focusing on the teeth.

The view transitions into a zoomed in human tooth, showing a clean cross-section. Inside the tooth, the pulp chamber is highlighted. The scene shifts to an animation of cells dividing, gently multiplying in a rhythmic, scientific style.

Next, clusters of stem cell like glowing spheres appear inside the pulp. These stem cell clusters pop out from the pulp chamber, floating upward. Each cluster then travels toward different parts of the human body shown earlier; symbolising how dental pulp stem cells can contribute to personalised treatment for many tissues.

The animation ends with the human body glowing softly, emphasising the regenerative potential of these cells.

Voice over script:

[Intro →Human body rotating] "What exactly are stem cells? Think of them as the body's master builders special cells with the unique ability to renew themselves and transform into many different types of tissues."

[Body parts highlighted: head, heart, bones, muscles, skin] "They play a role in how we grow, how we heal, and how our organs repair themselves. From the brain and heart to bones, muscles, and skin, stem cells are involved in almost every part of the body."

[Camera zooms into the mouth → tooth] "But one of the most surprising places we find them is inside our teeth."

[Tooth cross-section appears] "When we zoom into a tooth, we reach the pulp; the soft, living centre. This is where a special type of stem cell lives: **dental pulp stem cells**."

[Cells dividing rhythmically] "These cells can divide, renew, and multiply, just like other stem cells in the body."

[Glowing stem cell clusters appear in the pulp] "Inside the pulp, clusters of these powerful cells sit quietly, waiting for the right signals."

[Stem cells pop out and travel to different body parts] “for stem cell therapy dental pulp stem cells can be guided to become like bone cells, nerve cells, cartilage cells, and more. That’s why scientists are so interested in them, they have the potential to support healing and regeneration across many tissues.”

[Ending → human body glowing softly] “So when we talk about stem cells, we’re really talking about the body’s natural toolkit for repair. And inside every tooth, there’s a tiny reservoir of these remarkable cells; full of promise for future regenerative stem cell therapy.”

Content for Video 2

Introductory phrase: "Why Baby Teeth Are More Valuable Than Adult Teeth for Stem cell therapy".

Animation: The animation begins with a time-lapse of a child growing into an adult. As the character develops, simple labels appear on the side:

- Primary Dentition (Baby Teeth)
- Secondary Dentition (Adult Teeth)

The timeline shows the natural stages: an edentulous baby, the eruption of Baby Teeth, the transition phase where Baby Teeth fall out, and finally the full adult dentition.

The scene then shifts to a side-by-side comparison of two teeth:

- Left: Baby tooth
- Right: Adult tooth

On the Baby tooth side, the animation zooms into a child's mouth. The Baby tooth gently wobbles, showing a soft, natural "wiggle" motion. It then pops out smoothly with a cheerful bounce, emphasising natural shedding.

On the adult tooth side, the animation contrasts this by showing an adult tooth with visible decay or damage or a wisdom tooth. A dental extraction instrument appears briefly, indicating that adult teeth are often removed due to caries, disease and inflammation (pericoronitis). The extracted adult tooth is then shown as unsuitable and discarded, reinforcing the difference in sample quality.

The sequence ends with the Baby tooth highlighted as the naturally shed, healthy, stem cell therapy ready sample.

Voiceover Script:

[Intro → Time-lapse of child growing into adult] "Why are Baby Teeth more valuable than adult teeth for stem cell therapy? To understand this, it helps to look at how our teeth develop as we grow."

[Labels: Primary Dentition / Secondary Dentition] "We're born without teeth, then our first set — the primary or Baby Teeth begin to appear. These are naturally replaced as we grow older, making way for the permanent adult teeth."

[Timeline: baby → Baby Teeth → transition → adult teeth] "This natural cycle means that Baby Teeth fall out on their own, without disease, trauma, or dental treatment."

[Scene shifts to side-by-side comparison: Baby tooth vs adult tooth] “Now let’s compare a Baby tooth with an adult tooth.”

[Baby tooth zooms into child’s mouth, wobbles, pops out] “Baby Teeth loosen gently and shed naturally. They’re usually healthy, untouched by decay, and free from inflammation. When they pop out, they provide a clean, high-quality tissue sample ideal for stem cell therapy.”

[Adult tooth shows decay/damage, extraction tool appears] “Adult teeth, on the other hand, are often removed because of decay, infection, or gum disease. These conditions damage the pulp inside the tooth, reducing the quality of the stem cells.”

[Adult tooth shown as unsuitable and discarded] “Because of this, adult teeth are usually least favoured tissue for stem cell therapy compared to Baby Teeth”

[Ending →Baby tooth highlighted as stem cell therapy ready] “That’s why naturally shed Baby Teeth are so valuable. They provide healthy, uncontaminated tissue a perfect starting point for studying healing, regeneration of different body tissues, and future medical breakthroughs.”

Content for Video 3

Introductory phrase: "How the Tooth4Life collection Process Works"

Animation: A little child with a wobbly tooth smile as the tooth gently loosens. A small ticking clock appears above, showing the moment the tooth is ready to come out. The tooth pops out happily, turning into a smiling baby tooth character. The child hands the tooth to their parent by gently lifting the baby tooth character, who carefully places it into the Tooth Collection Kit. The kit is sealed, packaged by the parent and posts it in the post office/ post box. Once the collection kit is received at the post office it begins its journey in a royal mail van along a road, travelling through simple, friendly scenery before arriving at a building clearly labelled "Dundee Dental School."

Voiceover Script:

[Intro → Child with a wobbly tooth, gentle smile] "How does the Tooth4Life collection process work? It's simple, safe, and takes just a few moments at home."

[Tooth loosening, small ticking clock appears] "When a child has a wobbly tooth, it naturally loosens over time. A small ticking clock appears to show that moment when the tooth is finally ready to come out."

[Tooth pops out, becomes a smiling baby tooth character] "When it pops out, the tooth becomes a valuable asset, a reminder that this tiny tooth can make a big difference."

[Child hands tooth to parent] "The child hands the tooth to their parent, who gently places it into the Tooth Collection Kit provided by Tooth4Life."

[Parent places tooth into the kit, seals it] "The tooth goes into the sterile tube having a storage solution, the kit is sealed, and everything is packaged securely."

[Kit travelling along a road through friendly scenery] "Once it's ready, the kit begins its journey. It travels along a simple, friendly road, a visual path from your home to our Tooth4Life team."

[Kit arrives at building labelled 'Dundee Dental School'] "It arrives at Dundee Dental School, where the tooth is handled safely and ethically. Our scientists process the tooth by extracting the dental stem cells inside and developing personalised therapies to support healing and the treatment of various diseases."

[Ending →warm, reassuring tone] "With just one naturally shed tooth, your child contributes to stem cell therapy treatments that may shape the future of regenerative medicine. A small act, with meaningful impact."